3.3.1 Number of research papers published per teacher in the Journals as notified on UGC CARE list during the Academic Year 2023

S.No	Title of paper	Name of the author/s
1	Power Quality Improvement Using Rabbit Optimization FOPID Controlled Photovoltic Battery Power Hybrid Power Filter	Mr. Nirmal Kumar Pandey
2	Asymmetrical Interval Type-2 Fuzzy Logic Controller based MPPT For PV System Under Sudden Irradiance changes	Mr. Nirmal Kumar Pandey
3	Optimization of Mechanical Properties of Bamboo Fiber Reinforced Epoxy Hybrid Nano Composites By Response Surface methodology	Dr. AbhijeetGanguly
4	A Review on Inventory Model For Defective And Deteriorating Goods.	Mr. Abhishek Kumar Gupta
5	Case Study of Inventory Model For Deteriorating Items Under Rework And Shortage	Mr. Abhishek Kumar Gupta
6	A Review on Metrices Of Performance Evaluation of Various Supply Chain And Service Supply Chain In Service Sector	Mr.Revendra Kumar Deshmukh
7	Comparison of Performance Evaluation Of Transportation Service Providing Firms Using Hybrid Tops Is And Vikor	Mr.Revendra Kumar Deshmukh
8	Empirical Analysis of the Emerging Trends in the Photovoltaic Technologies of Condensed Matter Physics	Dr. Santosh Sharma

SANTOSH KUMAR SHARMA

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Available online 19 October 2023, 100508
In Press, Journal Pre-proof What's this?

Power Quality Improvement using Rabbit Optimization FOPID Controlled Photovoltaic- Battery Powered Hybrid Power Filter

Nirmal Pandey [a], Rupendra Kumar Pachauri [a], Sushabhan Choudhary [a], Ahmad Faiz Minai [b]

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Abstract

This paper presents the tuning of a fractional proportional controller (FOPI) using artificial rabbit optimization for the control of photovoltaic (PV) and battery power series active power filters along with a shunt passive filter which is known as a hybrid power filter. The PV power generation is combined into a series active filter to alleviate the problem of voltage sag and swell in the power system and also supply real power to the grid. Generally, the proportional controller is used to control the series active filter, but this is a linear controller and is not able to control the non-linear power system effectively. This problem is effectively controlled by non-linear control, i.e., a fractional-order proportional integral controller. Another problem in FOPI is finding the optimal value for proportional gain, integral gain, and fractional order. There are no standard rules for finding these values. This will make system design more complex. In this paper, artificial rabbit optimization is used to simplify the system design by finding optimal values of the FOPI controller for the series active filter and enhance the overall performance of the system. The MATLAB 2020b/Simulink environment is utilized to assess the recommended controller's performance. According to the experimental findings, the suggested system has a total harmonic distortion (THD) value for load voltage of 0.10% and a total harmonic distortion value for grid current of 1.87%. The proposed artificial-rabbit optimized FOIPI controller effectively controls the system against voltage sag, swell, and non-linear loads and also obeys IEEE power quality standards.

Introduction

The extensive use of power electronics-based systems has caused quality of power issues in the electrical grid. The harmonic issue at the common coupling point is being caused by this power electronics-based technology. This is because the present-day grid and other linked pieces of equipment have a single point of connection [1]. Load voltage is affected by quality of power issues such as voltage swell and sag when it is not maintained at a constant per-unit voltage level, which is necessary for the proper operation of the power

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Asymmetrical interval Type-2 Fuzzy logic controller based MPPT for PV system under sudden irradiance changes

Nirmal Kumar Pandey [a], Rupendra Kumar Pachauri [b], S. Choudhury [b], Raj Kumar Sahu [c]

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Abstract

For more than a decade, extensive research has been carried out on extensive methods for photovoltaic (PV) performance enhancement. The surrounding environmental conditions influence the performance of the PV system. The major environmental conditions, i.e., sun irradiance and temperature, are required to harvest maximum energy. One of the most well-known fields of research is power point tracking (MPPT) approaches, which have gained a lot of attention in terms of enhancing maximum power of PV system. Researchers have proposed a variety of topologies for harvesting the most available power namely, Perturb and Observe (P&O), Incremental Conductance (IC) and soft computing-based methods, namely Fuzzy logic control (FLC), Artificial Neural networks (ANN), Adaptive Fuzzy Neural Interface (ANFIS), and metaheuristics. In this work, realistic solar irradiance, such as sudden and abrupt irradiance conditions, are created to investigate PV system performance for extensive analysis in terms of transient studies. For comparative study, an asymmetrical interval Type-2 FLC in addition to conventional P&O is being proposed for such realistic irradiance conditions. The performance indices such as global maximum power point (GMPP) location, transient behavior, and efficiency are studied to establish its superiority under the MATLAB/ Simulink environment.

Introduction

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ORIGINAL PAPER

Check for updates

Optimization of mechanical properties of bamboo fiber reinforced epoxy hybrid nano composites by response surface methodology

Roopesh Kumar[1,2] · Abhijeet Ganguly[1,2] · Rajesh Purohit[3]

Received: 28 September 2022 / Accepted: 19 January 2023

Abstract
At present, materials obtained from nature are adopted with high priority due to exploitation of natural resources ofthe materials. This work is focused on the use of natural fibre with nano-silica as reinforcement in epoxy resin as a matrix. The polymer composites were developed by mixing an appropriate amount of nano SiO_2 with bamboo fibres. After composite fabrication, specimens of standard size were prepared, and tests related to mechanical properties were performed. 32H compositesperformed best in the tensile test. The flexural test value for 32G composite was the highest. We found that the 32H composite had better energy absorption capacity. Response surface methodology (RSM) was used to find the optimum composition of composites, and the effects of fibre and nano-SiO_2 on their mechanical properties were investigated. A central composite design was employed to analyse the composite properties. A second order polynomial model was used for predicting strength of the composites. It has been found that the composite was best fit by a quadratic regression model with an excessive co-efficient to determine the R^2 value. Effects of bamboo fibre and nano-SiO_2 were examined using analysis of variance (ANOVA).Experiment found that two-layer natural bamboo fibre with 2 wt.% of silica is of high quality. Nano composites of fabricated natural fibre reinforced polymer has numerous uses in automotive, aircraft, aerospace, sporting, structural, and home appliance industries.

Keywords Bamboo fibers · Bidirectional woven mat · Nano SiO_2 · Response surface methodology (RSM) · ANOVA · Mechanical properties

1 Introduction

1.1 Physical, morphology and chemical properties

It is easier to maintainenvironmental balanceusing natural materials. The properties of the composites made are better than other materials being used.Many researchers have presented their work through different research articles on natural fibre composites, their applications, comparisons, and properties. According to Huaet al. composite materials offer excellent mechanical, thermal and tribological properties compared to other materials.Mechanical properties were improved usingepoxy with natural fibers [1]. Banga et al. [2] used epoxy resin CY-230 and hardener HY- 951 to make the bamboo fiberhybrid composite. The author used various weight percentages of the bamboo fiber to make four different composites 0, 20, 30 and 40wt % in the polymer matrix. Hand lay-up techniques were used to make composites. Interpreted the properties of material through mechanical and thermal properties tests as well as water absorption tests. Mohan et al. used natural fibres, viz. sisal and bamboo, for making the composites by hand lay-up technique. Explained advancement of the material according to current scenario. They reported that synthetic and natural fibre are same with respect to their material density, and both are nontoxic and nonabrasive [3]. Jain et al. performed experiments on bamboo fibres and their composites and tested the properties

✉ Roopesh Kumar
roopeshsinha12@gmail.com

Abhijeet Ganguly
abhijeetganguly@csitdurg.in

Rajesh Purohit
rpurohit73@gmail.com

1 Department of Mechanical Engineering, CSIT, Durg 491001, Chhattisgarh, India

2 Chhattisgarh Swami Vivekananda Technical University, Bhilai 491107, CG, India

3 Department of Mechanical Engineering, MANIT Bhopal, Bhopal, India

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International Journal for Scientific Research in Modern Engineering and Science

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A REVIEW ON INVENTORY MODEL FOR DEFECTIVE AND DETERIORATING GOODS

Praful Kumar Sahu [1]*, **Abhishek Kumar Gupta** [2]

[1] *M. Tech. Scholar, Department of Mechanical Engineering, CSIT Durg, C.G., India.*
[2] *Assistant Professor, Department of Mechanical Engineering, CSIT Durg, C.G., India.*

Abstract

Inventory control can be applied for every industrial and non industrial application, but specifically when apply to did you reading and defected item it is very helpful in managing inventory of wastage. Also it helps in recycling and restoring the equipment which is mainly considered as scrap in many industries. Considering environmental aspects, it save the resources and it optimise the use of inventory which reduce the overall cost of production. This literature review consist of study of various item in industries such as raw material, spare/ repair parts, intermediate goods and finished goods. This re is done for deteorating items and defective item goods under various conditions which allow the marked down policy of company, goods allowed with inflation, goods allowed with preservation technology, goods allowed allowed with three level production setup, inventory model allowed with single vendor and single buyer, inventory model allowed with shortages.

The finding of the research or literature review is that inventory control for deteorating item and re manufacturing items under different condition need to be analysed. Concluding, inventory control model or case study for various condition such as backlogging, inflation, markdown policy, variable demand, multiple production system, preservation technology can be analysed.

Keywords: EPQ, ELDSP, EMQ, MRP, EOQ, Inventory Control, Remanufacturing Goods.

* *Corresponding author*

1. Inventory Control

Inventory is coordination and supervision of the supply, storage, distribution, and recording of materials to maintain quantities adequate for current needs without excessive oversupply or loss. The inventory is used to represent the aggregate of those items of tangible assets which are:

Held for sale in ordinary course of the business.

In process of production for such sale.

To be currently consumed in the production of goods or services to be available for sale.

The inventory may be classified into three categories:

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CASE STUDY OF INVENTORY MODEL FOR DETERIORATING ITEMS UNDER REWORK AND SHORTAGE

Praful Kumar Sahu [1]*, Abhishek Kumar Gupta[2]

[1]*M. Tech. Scholar, Department of Mechanical Engineering, CSIT Durg, C.G., India.*
[2]*Assistant Professor, Department of Mechanical Engineering, CSIT Durg, C.G., India.*

Abstract

Inventory management is an important part in any industry which have inventory in the form of raw material, finished good and semifinished goods. Mini companies can save large amount of money by using inventory management techniques. It is important for every company to reduce the loss and keep the cost of finished good very competitive but if we consider deteriorating item, it is very necessary to rework on it so as to save various cost associated with it. Also continuous supply of material, smooth production, change in prices, stock out, minimising inventory cost, lot purchase are the function of inventory cost.

Various literature related to inventory control have been studied which determines in actual practice fabrication of defective items is to be anticipated due to the deterioration process. The EPQ model problem for single item under the assumption of imperfect production and perfect work have been broadly carrie out in past decades. In real manufacturing process, quality of rework is always imperfect. Various items which comes under deteriorating inventory model are electronic products, fashion clothing, pharmaceuticals, paper based material, foods, vegetable, fruits and chemical.

Case study is carried out in BEC Foods and Private Limited, Kuthrel Durg (C.G.). Objective of the research is to analyse and create an inventory model under consideration of failure rework can shortages. Also to find optimum replacement runtime and perform sensitivity analysis for the case study.

Methodology involves mathematical formulation which consists of calculation of various Time, Total cost and Inventory Size associated with inventory model under rework and shortages. At every steps while calculating the optimum cost, rework time the condition of optimum inventory also need to be calculated. Also Sensitivity analysis is carried out by changing the value of penalty cost of selling deterioted items, production setup cost, shortage production setup cost, serviceable item holding cost, defective item holding cost. Optimum solution of inventory model is shown in table in the paper which concludes that increase in penalty cost lead to increase in optimal shortage accumulation time, optimal complete backlogging time and minimum total cost. Optimal replinishment runtime, optimal rework run time, optimum non production runtime, optimum replenishment cycle time and optimum economic production quantity decreases. Similarly sensitivity analysis is done for other factor also.

23

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International Journal for Scientific Research in Modern Engineering and Science

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A REVIEW ON METRICES OF PERFORMANCE EVALUATION OF VARIOUS SUPPLY CHAIN AND SERVICE SUPPLY CHAIN IN SERVICE SECTOR

Mehandra Pal Damle [1]*, Revendra Kumar Deshmukh [2]

[1] *M. Tech. Student, Department of Mechanical Engineering, CSIT Durg, (C.G.), India.*

[2] *Assistant Professor, Department of Mechanical Engineering, CSIT Durg (C.G.), India.*

Abstract

Performance evaluation is a key activity of supply chain in the dynamic global competitive scenario for the improvement of productivity and profitability. As there is enough scope for improvement in productivity and profitability, supply chain managers need to strive for collection, analysis, and interpretation of qualitative and quantitative information to measure and compare in order to give the right direction to enhance supply chain performance. Then a comprehensive performance evaluation system needs to be developed and redefined to monitor, control, and direct total supply chain operations on a continuous basis by incorporating the entire supply chain process as an integrated system.

Service sector is becoming as a lifeline for the social and economic growth of any country. It is well known that contribution of service sector to nation's progress is substantial. Services contribute twice the economic output compared with manufacturing. Evaluating service supply chains is essential to measure the growth. Very little attention has been paid to performance evaluation of service supply chains and hence there is a pressing need to direct research efforts in this direction. Since the output of service is intangible, heterogeneous and simultaneous, identifying suitable evaluation criteria is a crucial exercise. Factors like, the nature of the service industry the firms under evaluation belong to and the country and region where they are operating, can influence the list of evaluation criteria.

Various research paper have been studied and came to the conclusion that List of performance metrics applicable for evaluating performance and Multi criteria decision making approaches appropriate for carrying out performance evaluation of service supply chains need to be identified. Also the suitability, effectiveness and applicability of the approaches recommended for performance evaluation of service supply chains need to be demonstrated by carrying out service supply chain case studies in an emerging economy.

Keywords: SCM, DEA, MCDM, TOPSIS, VIKOR, KPI.

* *Corresponding author*

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COMPARISON OF PERFORMANCE EVALUATION OF TRANSPORTATION SERVICE PROVIDING FIRMS USING HYBRID TOPSIS AND VIKOR

Mehandra Pal Damle [1]*, Revendra Kumar Deshmukh [2]

[1]*M. Tech. Student, Department of Mechanical Engineering, CSIT Durg, (C.G.), India.*

[2]*Assistant Professor, Department of Mechanical Engineering, CSIT Durg (C.G.), India.*

Abstract

Performance evaluation is a key activity of supply chain in the dynamic global competitive scenario for the improvement of productivity and profitability. As there is enough scope for improvement in productivity and profitability, supply chain managers need to strive for collection, analysis, and interpretation of qualitative and quantitative information to measure and compare in order to give the right direction to enhance supply chain performance. Then a comprehensive performance evaluation system needs to be developed and redefined to monitor, control, and direct total supply chain operations on a continuous basis by incorporating the entire supply chain process as an integrated system.

Service sector is becoming as a lifeline for the social and economic growth of any country. Evaluating service supply chains is essential to measure the growth. Very little attention has been paid to performance evaluation of service supply chains and hence there is a pressing need to direct research efforts in this direction. Since the output of service is intangible, heterogeneous and simultaneous, identifying suitable evaluation criteria is a crucial exercise.

In this work, a comprehensive listing of metrics suitable for performance evaluation of service supply chains was brought out. Performance evaluation of supply chains and the process of decision making based on the outcome of the evaluation is a multi criteria decision making (MCDM) process. As part of this research work, two hybrid multi criteria decision making approaches have been proposed for carrying out the evaluation of comparative performances of service supply chains. The fuzzy TOPSIS approach and the fuzzy VIKOR approach have been implemented for evaluating the performances of transportation service providing (TSP) firms. Comparison of ranking is been done in this paper. The approaches proposed here, once incorporated and institutionalized into the organizations can be very effective for practicing executives of organizations to evaluate and monitor the performances of supply chains employed by the organizations. The approaches are simple to learn and implement. The procedural steps are less time consuming both with or without the use of computers. The approaches are free from accusations of bias and they are very much suitable for standardization.

Keywords: SCM, DEA, MCDM, HYBRID TOPSIS, HYBRID VIKOR, KPI, SCOR, TSP.

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Том 15 № 4, 04009(6cc) (2023)

Empirical Analysis of the Emerging Trends in the Photovoltaic Technologies of Condensed Matter Physics

Neeta Deshpande[1,*], Santosh Kumar Sharma[2], P. William[3], A.B. Pawar[4]

[1] *Department of Computer Engineering, R H Sapat College of Engineering, Management Studies and Research, SPPU, Pune, India*
[2] *Department of Mechanical Engineering, Chhattrapati Shivaji Institute of Technology, CSVTU, Bhilai*
[3] *Department of Information Technology, Sanjivani College of Engineering, SPPU, Pune, India*
[4] *Department of Computer Engineering, Sanjivani College of Engineering, SPPU, Pune, India*

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We examine some current developments in a subject that is developing at the intersection of conventional inorganic and organic materials. Inorganic and organic structural components co-exist in crystalline systems that make up hybrid inorganic-organic framework materials. Porous hybrid frameworks have established a lot of concentration in this field during the last several years because of their potential use in catalysis, separations, and sensors. Our study primarily focuses on the magnetic, optical, electrical, and dielectric characteristics that are normally the preview of condensed matter physics. The value of empirical research in understanding the processes of social, economic, and technological development cannot be overstated. In recent years, there has also been an increase in the deposition of thin films of hybrid compounds onto solid surfaces for possible use in surface chemistry and physics. The development of new technologies, like quantum computing and spintronic, and the understanding of how matter behaves under extreme situations, we demonstrate that these materials display a great variety of behavior in these domains and provide several fascinating prospects to the physics community. We also provide a brief overview of several of the characteristics of porous materials in terms of nano technology with hybrid approach. The field of hybrid technology still has a critical need for theory and simulation.

Keywords: Condensed Matter Physics, Inorganic and Organic Materials, Electronic and Dielectric Properties, Magnetism, Optical.

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PACS numbers: 77.84.Jd, 88.30.rf

1. INTRODUCTION

The empirical analysis is an approach of investigation that gathers and examines information from studies or observations to draw inferences or test hypotheses. It is a methodical technique for learning about the world that utilizes data acquired through direct or indirect observations instead of theory, speculative ideas, or presumptions. To conduct an empirical analysis, data must be gathered from a variety of sources, including surveys, interviews, experiments, and observational studies. After then, the data is statistically analyzed to find any noteworthy trends, patterns, or connections. Researchers and decision-makers may better grasp the possibilities and problems confronting their disciplines and create efficient plans to address them by recognizing and evaluating emerging trends [1]. Numerous modeling and empirical techniques have been used in a large and expanding body of literature to estimate the direct and indirect economic effects of natural disasters. The new body of research synthesizes its key theoretical, computational, and empirical results, and discusses insights into the elements and practices that have been discovered to reduce the effects of disasters to close this gap [2]. Photovoltaic technologies refer to the conversion of sunlight into electrical energy using solar cells made of semiconductor materials such as silicon. Solar cells are the building blocks of PV modules, which are the primary components of solar panels used in solar power systems. Numerous benefits of photovoltaic technologies include their ability to produce electricity without emitting greenhouse gases, their minimal maintenance needs, and their long lifespans. Large-scale solar power plants as well as grid-connected and off-grid solar power systems for residential, commercial, and industrial purposes utilize them regularly [3]. Condensed matter physics is the branch of physics that studies the physical properties of matter in its condensed state, which includes solids and liquids. It deals with the behavior of large groups of particles, such as atoms, molecules, and electrons, and how they interact to produce the macroscopic properties of matter, such as elasticity, thermal conductivity, and electrical conductivity. Several sectors, including materials science, electronics, and nanotechnology, use condensed matter physics. The development of new technologies, like quantum computing and spintronic, and the understanding of how matter behaves under extreme situations, such as high pressure and low temperature, depend on this knowledge [4].

* neeta.deshpande@ges-coengg.org

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